JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

AUGUST 2022 EXAMINATIONS

OPTION – C101

CHEMISTRY **Time Allowed: 3 hours**

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer ALL questions in this section.

Use the OMR answer sheet provided to answer the questions.

Write and shade on your OMR Sheet the OPTION written on your question paper.

Follow the instructions on the OMR sheet.

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in ALL; ONE Question from each Course.

Ensure you read and follow all the Instructions on the cover page of the Answer Booklet.

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer ALL Questions.

1. Which of the following information is obtained from the spectrum of a spectrometer?
   A. Mass / charge ratio.
   B. Charge / mass ratio.
   C. Atomic mass ratio.
   D. Charge ratio.

2. Identify a pair of compound with the same empirical formula in the options below.
   A. C_2H_2 and C_6H_6
   B. C_3H_6 and C_4H_{10}
   C. NO_2 and N_2O_5
   D. C_6H_5OH and $C_{12}H_{10}O$

3. Which of the following pairs of elements will react together to form a compound with ionic bonding?
   A. B and H
   B. Rb and Cl
   C. S and S
   D. C and F

4. Which of the following is **NOT** a unit of measurement of volume?
   A. Quart.
   B. Ounce.
   C. Cubic metre.
   D. Litre.

5. Which of the following is **NOT** a shape obtained using VSEPR?
   A. Trigonal bipyramidal.
   B. Trigonal planar.
   C. Tetragonal pyramidal.
   D. Square planar.

6. A student obtained the following values from the measurement of a physical quantity. Which of the measurements is the most precise?

A. 4.0000

B. 4.0

C. 4.000

D. 4.00

7. A measuring cylinder labeled 25 mL has its actual volume to be 24 mL, what type of error(s) will the cylinder introduce into measurements?

A. Random error.

B. Random and systemic error.

C. Systemic error.

D. Error due to parallax.

8. The atomic structure described as a diffuse cloud of positive charge with negative electrons embedded in it is known as

A. Alpha scattering model.

B. Plum pudding model.

C. Bohr model.

D. Rutherford's model.

9. The purpose of heat exchanger in the Haber process is to

A. cool the incoming gas mixture to avoid over heating the catalyst.

B. cool the reaction product and separate NH_3 from the unused N_2 and H_2.

C. warm the incoming gas mixture and shift the equilibrium to give more NH_3.

D. warm the incoming gas mixture and speed up the reaction.

10. What is the wavelength of the radio-waves with a frequency of 12 MHz ?

$(c = 3.0 \times 10^8\ ms^{-1})$

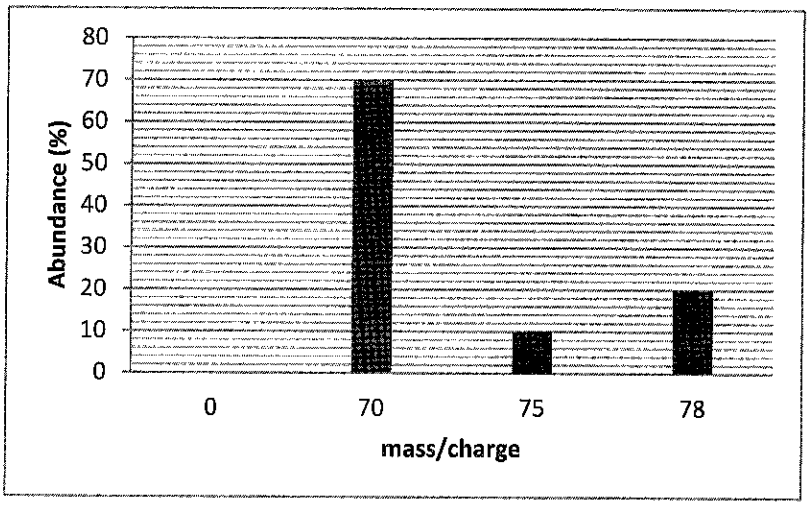
A. 0.25 m

B. 2.50 m

C. 25.0 m

D. 250.0 m

11. The mass spectrum of an unknown element is displaced below. The atomic mass of the unknown element is



A. 70

B. 75

C. 78

D. 72

12. Which of the following statements can be attributed to Niels Bohr?

i. Electrons in a hydrogen atom moves around the nucleus only in certain allowed circular orbits.

ii. There is a fundamental limitation to just how precisely we can know both the position and momentum of a particle at a given time.

iii. In a given atom, no two electrons can have the same set of quantum numbers.

A. i.

B. ii

C. iii.

D. i. and ii.

13. Iodine trichloride has 2 lone pairs and 3 bond pairs. Predict a shape for the molecule.

A. T- shaped.

B. Trigonal pyramidal.

C. Tetrahedral.

D. Trigonal bipyramidal.

14. The general trend of atomic radius and ionization energy follow the same trend i.e. decrease across the period and increases down the group. This means that
    A. it is easier to remove an electron across a period.
    B. each successive element down a group has electrons in a higher energy level.
    C. each successive element across a period has electrons in a higher energy level.
    D. it is difficult to remove an electron down the group.

15. Which reaction corresponds to ΔH_f for $Na_2CO_{3\,(s)}$ at 298K?
    A. $2Na_{(s)} + C_{(s)} + {}^3/_2O_{2\,(g)} \longrightarrow Na_2CO_{3(s)}$
    B. $Na_2O_{(s)} + CO_{2\,(g)} \longrightarrow Na_2CO_{3(s)}$
    C. $2Na^{+}{}_{(aq)} + CO_3^{2-}{}_{(aq)} \longrightarrow Na_2CO_{3(s)}$
    D. $2Na^{+}{}_{(aq)} + OH^{-}{}_{(aq)} + CO_{2\,(g)} \longrightarrow Na_2CO_{3(s)} + H_2O_{(l)}$

16. What will be the effect of an increase in temperature on the following reaction?

    $H_3PO_4\,(aq) + H_2O\,(l) \rightleftharpoons H_2PO_4^-(aq) \qquad \Delta H = -13KJ$ at 298K
    A. favour the forward reaction.
    B. have no effect on the reaction.
    C. favour the reverse reaction.
    D. slow down the reverse reaction.

17. Liquefaction occurs when gases are subjected to
    A. low pressure and low temperature.
    B. low pressure and high temperature.
    C. high pressure and low temperature.
    D. high pressure and high temperature.

18. Which of the following is transferred from cathode to anode through a membrane in hydrogen fuel cell?
    A. Proton.
    B. Electron.
    C. Neutron.
    D. Hydrogen gas.

19. Identify the rate-determining step in the reaction A $\longrightarrow$ E, which goes through the following steps:

$$A \xrightarrow[1.2\ secs]{} B \xrightarrow[56\ secs]{} C \xrightarrow[2.1\ secs]{} D \xrightarrow[1.55\ secs]{} E$$

A. A $\longrightarrow$ B

B. B $\longrightarrow$ C

C. C $\longrightarrow$ D

D. D $\longrightarrow$ E

20. What is the conjugate base in the reaction of bicarbonate ion, HCO_3^- with water.

$$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$$

A. HCO_3^-

B. H_2O

C. H_2CO_3

D. OH^-

21. Which of the following statement is **NOT** true about the effect of an increase in the temperature on the distribution of molecular speeds in a gas?

A. The most probable speed increases.

B. The collision of the molecule increases.

C. The area under the distribution curve becomes broader.

D. The area under the distribution curve remains the same.

22. A mixture of gases contained in a vessel of 1.5 atm is found to be 15 % N_2, 50 % of N_2O and 35 % CO_2 by volume. Calculate the partial pressure of nitrogen gas. (N = 14, C = 12, O = 16).

A. 0.75 atm.

B. 0.98 atm.

C. 0.23 atm.

D. 1.28 atm.

23. In which reaction will the pressure equilibrium constant (K_p) equals the concentration equilibrium constant (K_c)?

A. $PCl_5\ (g) \rightleftharpoons PCl_3\ (g) + Cl_2\ (g)$
B. $2H_2O\ (g) \rightleftharpoons H_2\ (g) + O_2\ (g)$
C. $2SO_2\ (g) + O_2\ (g) \rightleftharpoons 2SO_3\ (g)$
D. $N_2\ (g) + 3H_2\ (g) \rightleftharpoons 2NH_3\ (g)$

24. When the half life of a reaction is inversely proportional to the initial concentration of the reactant, the kinetics follow

A. zero order.
B. first order.
C. second order.
D. third order.

25. Which pair of gases has the same average rate of diffusion at 25°C? [C = 12, N = 14, O = 16].

A. N_2O and CO_2
B. N_2 and O_2
C. NO and CO
D. NO_2 and CO_2

26. The following are true of boron **EXCEPT**

A. it is a non-metal.
B. it is a typical metal.
C. it exhibits covalency.
D. it occurs in two isotopic form.

27. The following describe the properties of transition element **EXCEPT**

A. formation of coloured ions.
B. constant oxidation state.
C. formation of complex ions.
D. Catalytic ability.

28. The following ions are monodentate ligands **EXCEPT**

A. Cyanide.

B. Thiocyanate.

C. Bromide.

D. Oxalate.

29. Graphite has a large network of strong covalent bonds, which of the following is **NOT** a property of graphite.

A. High melting point.

B. Electrical Conductivity.

C. Presence of sp^2 hybridised carbon atoms.

D. Presence of sp^3 hybridised carbon atoms.

30. The correct order of melting point of B, Al and Ga is

A. B > Al > Ga

B. Al > B > Ga

C. Ga > B > Al

D. B > Ga > Al

31. An element with high electronnegativity has

A. high Ionization Energy and high Electron Affinity.

B. high Ionization Energy and low Electron Affinity.

C. low Ionization Energy and high Electron Affinity.

D. low Ionization Energy and low Electron Affinity.

32. Which statement about the reactions of lithium or its compounds is based on the diagonal relationship with magnesium?

A. Lithium burns very slowly in oxygen.

B. Lithium carbonate decomposes readily on heating, forming Li_2O, and CO_2.

C. Lithium nitrate decomposes on heating, forming lithium nitrite, and oxygen.

D. Lithium reacts very violently with cold water, producing hydrogen.

33. The coordination number of $[Co(en)(H_2O)_4]Cl_3$ complex is

A. 4

B. 5

C. 6

D. 8

34. The condensed electron configuration of Chloride ion is

A. [Ne] $3s^1\ 3p^6$

B. [Ne] $3s^2\ 3p^6$

C. [Ne] $3s^2\ 3p^5$

D. [Ne] $3s^2\ 3p^4$

35. Why do lithium halides have greater degree of covalent character than halides of other members of the group?

A. The small size of Li^+ gives it a high polarizing power which leads to covalency.

B. The small charge of Li^+ gives it a low polarizing power and high covalent character.

C. Li^+ is the most electronegative and confers higher covalent character.

D. Li^+ has equivalent size with halide ions and greater covalent character.

36. List the group 1 elements, K, Na, Li, Rb, Cs in order of increasing reactivity with water.

A. Na < Li < Rb < K < Cs

B. K < Rb < Cs < Na < Li

C. Rb < Cs < Na < K < Li

D. Li < Na < K < Rb < Cs

37. In which of the following reactions does chlorine disproportionate?

A. $AgNO_3 + HCl \longrightarrow AgCl + HNO_3$

B. $Cl_2 + H_2O \longrightarrow HCl + HOCl$

C. $2KClO_3 \longrightarrow 2KCl + 3O_2$

D. $HCl + NaOH \longrightarrow NaCl + H_2O$

38. Carbon in its excited state has which of the following electronic configuration?

A. $1s^2\ 2s^2\ 2p^2$

B. $1s^1\ 2s^2\ 2p^3$

C. $1s^2\ 2s^1\ 2p^3$

D. $1s^1\ 2s^1\ 2p^3$

39. The advantage of biotechnology as a tool for plant breeding over other plant breeding techniques is that it

A. is less precise and produces specific gene to make changes.

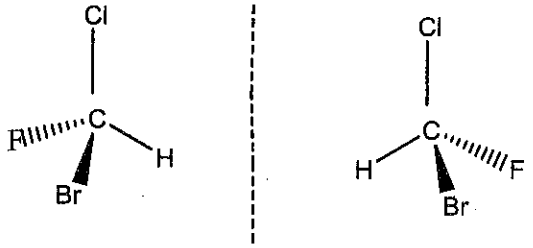
B. is more precise and isolates specific gene to make exact changes.

C. is precise and makes definite gene to particular changes.

D. isolates gene with certain characteristics.

40. Concentrated tetraoxosulphate (VI) acid in the reaction of ethanol and ethanoic acid acts as
    A. a solvent.
    B. a catalyst.
    C. an oxidizing agent.
    D. a dehydrating agent.

41. The following molecules are



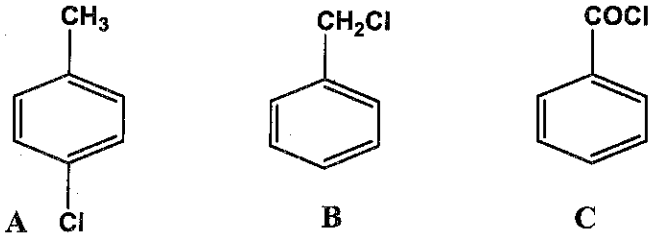
    A. Geometrical isomers.
    B. Diastereomers.
    C. Optical isomers.
    D. Tautomers.

42. The respective stationary phase in the column chromatography and paper chromatography are usually
    A. SnO_2 and SiO_2
    B. Paper and SiO_2
    C. SiO_2 and paper
    D. Al_2O_3 and SiO_2

43. The two species formed in the homolytic fission of Cl-Cl bond are called
    A. ions.
    B. molecules.
    C. free radicals.
    D. hybrids.

44. The boiling point of butanoic acid is higher than ethyl ethanoate despite being isomers, this can be attributed to
    A. hydrogen bond formed in butanoic acid.
    B. ethyl ethanoate being an ester.
    C. the inductive effect of the alkyl group.
    D. the steric hindrance of the alkyl group on ethyl ethanoate.

45. The three chlorinated benzene derivatives A, B and C vary in their ease of hydrolysis



Choose the order of the ease of hydrolysis

A. A > B > C

B. C > B > A

C. A > C > B

D. C > A > B

46. The balanced equation when CH_3COCl reacts with ethanol to produce a fruity – smelling liquid is

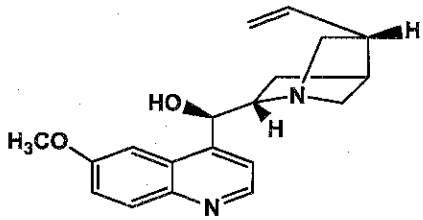
A. $CH_3COCl\ (aq) + CH_3CH_2OH\ (aq) \longrightarrow CH_3CH_2OCH_2CH_3 + HCl$

B. $CH_3COCl\ (aq) + CH_3CH_2OH\ (aq) \longrightarrow CH_3COOCH_2CH_3 + HCl$

C. $CH_3COCl\ (aq) + CH_3CH_2OH\ (aq) \longrightarrow CH_3COCH_2CH_2OH + HCl$

D. $CH_3COCl\ (aq) + CH_3CH_2OH\ (aq) \longrightarrow CH_3CH_2COCH_2OH + HCl$

47. The organic compound, **Quinine**, is a secondary metabolite (natural product) isolated from a medicinal plant, which possesses antimalarial properties. Classify the hydroxyl group in Quinine.



Quinine

A. Primary alcohol.

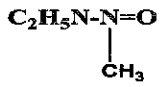
B. Tertiary alcohol.

C. Secondary alcohol.

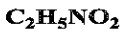
D. Quartenary alcohol.

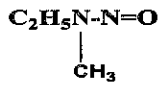
48. Deduce X in the reaction below



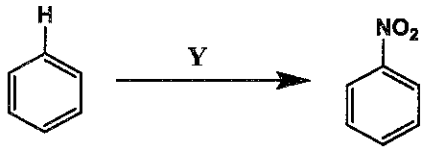
A. 

B. 

C. 

D 

49. Identify Y in the reaction below, and the type of reaction it is termed



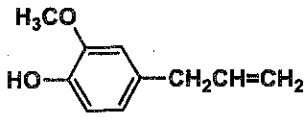
A. Conc. HNO_3, substitution reaction.

B. Conc. HNO_3 / H_2SO_4, addition reaction.

C. Conc. HNO_3 / H_2SO_4, electrophilic substitutution reaction.

D. Conc. H_2SO_4, electrophilic substitutution reaction.

50. What kind of polymerization can occur with eugenol which has the structure below?



A. Condensation polymerization

B. Geometrical polymerization

C. Optical polymerization

D. Addition polymerization

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in ALL; ONE Question from each Course.

CHM 001: GENERAL CHEMISTRY

1. (a) (i) State the Periodic law.

(ii) State the **group, period** and **block** of the elements calcium ($_{20}Ca$) and Bromine ($_{35}Br$).

[3½ Marks]

(b) 4.0 g of NaCl is heated with manganese (IV) oxide in excess concentrated tetraoxosulphate (IV) acid, the chlorine gas liberated is then passed through KI solution to liberate iodine gas.

(i) Balance the equation for the reaction.

$$NaCl + H_2SO_4 + MnO_2 \rightarrow MnSO_4 + NaHSO_4 + H_2O + Cl_2$$

(ii) Calculate the mass of MnO_2 used.

(iii) Provide an equation for the production of potassium chloride from the reaction of chlorine and potassium iodide.

(iv) Calculate the mass of iodine liberated.

[Na= 23, Cl = 35.5, H = 1, S = 32, Mn = 55, O = 16, I = 127] **[3½ Marks]**

(c) (i) What is a precipitation reaction and why do precipitates form?
Write balanced molecular and net ionic equations for each of the following

(ii) $NH_4Cl_{(aq)} + KOH_{(aq)} \rightarrow$

(iii) $Al(NO_3)_{3(aq)} + Ba(OH)_{2(aq)} \rightarrow$ **[3 Marks]**

2. (a) (i) Define the term 'electronegativity'.

(ii) Write the Lewis structure for the following compounds: ICl_4^- *and* PCl_3

(iii) Explain the difference between using Lewis structures and VSEPR to provide the shapes/structures of molecules. **[3 Marks]**

(b) (i) Determine the empirical formula of a compound used as explosive which contains C, H, N and O, if 50 g of the compound was found to contain 31.72 g of O, 7.93 g of C and 1.10 g of H. [M_r of explosive = 227 g]

(ii) Write an equation for the decomposition of the explosive obtained in **(b) (i)** if it decomposes into carbon dioxide, nitrogen, oxygen and water. **[4 Marks]**

(c) Percentages of protein content in milk were 54.01, 54.24, 54.05, 54.27 and 54.11. The accepted true value is 54.20. Calculate:

(i) Absolute error.

(ii) Relative error.

(iii) Mean Deviation.

(iv) Standard Deviation. **[3 Marks]**

3. (a) (i) Define buffer solution.
   (ii) Write the Ksp expressions for saturated solutions of AgCl and $Fe(OH)_2$. **[3 Marks]**

   (b) (i) Explain the term electrochemical cells.
   (ii) Using equation only, explain the electrochemical process involved in the rusting of iron
   (iii) Obtain a formula for the Nernst equation from the free energy in a non-standard state to one in a standard state. **[3 Marks]**

   (c) (i) How does neutron to proton ratio affect stability of a radioactive element?
   (ii) Positron emission from ^{11}C occurs with release of $3.45 \times 10^{11} Jmol^{-1}$ energy. What is the mass per mole of ^{11}C in the following nuclear reaction?
   $^{11}_{6}C \rightarrow {}^{11}_{5}B + {}^{0}_{1}e$ $\quad [c = 3 \times 10^8\ ms^{-1}]$ **[2 Marks]**

   (d) What is the standard enthalpy of formation of ethene, (C_2H_4), if the standard enthalpy of combustion is $-1411\ kJmol^{-1}$?
   $\Delta H^o_{f\,CO_2} = -394\ kJmol^{-1}$
   $\Delta H^o_{f H_2O} = -286\ kJmol^{-1}$ **[2 Marks]**

4. (a) State Faraday's second law of electrolysis. **[1 Mark]**

   (b) A direct current of 200 mA flows for 3 hours through three cells in series. They contain solutions of sodium chloride, copper(II) tetraoxosulphate(VI) and gold(III) trioxonitrate(V). Calculate the mass of metal deposited in each cell.
   [Na = 23.0, Cu = 63.5, Au = 197] **[3 Marks]**

   (c) (i) Define solubility product.
   (ii) State **THREE** factors that affect solubility product.
   (iii) In the equation below

   $$Ag_2CrO_{4(s)} \rightleftharpoons 2Ag^+_{(aq)} + CrO_4^{2-}{}_{(aq)}$$

   Calculate the solubility product (Ksp) of Ag_2CrO_4 if the concentration of Ag^+ is $1.5 \times 10^{-4} M$ **[3 Marks]**

   (d) (i) State **TWO** uses of radioactive isotopes.
   (ii) The half – life of the isotope of uranium of mass number 234g is 3.5×10^5 years. How long after the isolation of a sample of this isotope will only one fifth of the original mass be left? **[3 Marks]**

CHM 003: INORGANIC CHEMISTRY

5. (a) Outline **FOUR** differences between the most popular allotropes of carbon. **[2 Marks]**

(b) Give reasons for the following observations:

(i) Group 2A elements always shows small positive electron affinity.

(ii) Beryllium ion is amphoteric but magnesium ion is ionic.

(iii) Non-metallic properties of elements increases across the period.

(iv) Group IIA element always form colourless compound. **[2 Marks]**

(c) Give the name of the following complexes;

(*i*) $K_4[Fe(CN)_6]$

(*ii*) $[Co(SO_4)_2Cl_2]^{4-}$

(*iii*) $[Ni(NH_2CH_2CH_2NH_2)_2Cl_2]^{2+}$ **[3 Marks]**

(d) Show how the combustion of sulphur-containing fossil fuel causes acid rain in a polluted industrialized environment. (Name all the products formed) **[3 Marks]**

6. (a) The properties of lithium are quite different from the properties of other members of group one elements but similar to those of magnesium.
State **SIX** similarities between lithium and magnesium. **[3 Marks]**

(b) (i) List the **TWO** major gases which cause acid rain and give the non-natural (artificial) sources of these gases.

(ii) What products are formed when each of the following elements reacts with red phosphorous?

I. Bromine

II. Chlorine **[4 Marks]**

(c) (i) Give reason why compounds of Sc^{3+} and Zn^{2+} are mostly white whereas those of Sc^{2+} and Cu^{2+} are coloured.

(ii) List **TWO** distinct features of these transition metals. **[3 Marks]**

CHM 004: ORGANIC CHEMISTRY

7. (a) An alcohol has the molecular formula C_3H_8O. When warmed with an alkaline solution of iodine it forms a yellow precipitate.

(i) Give the reaction scheme and name the yellow precipitate.

(ii) Draw the displayed formula of the alcohol.

(iii) The first stage in the reaction of the alcohol with alkaline iodine solution is an oxidation reaction.
Name the organic product of this first stage. **[3 Marks]**

(b) A hydrocarbon X, contains 92.3% carbon by mass. The relative molecular mass of X is 78.0

(i) Calculate the empirical formula of X.

(ii) Calculate the molecular formula of X. **[4 Marks]**

(c) (i) Draw the structures of the **TWO** geometrical isomers of 1, 2 – dibromoethane.

(ii) What feature of the double bond prevents isomer 1 from changing into isomer 2?

(iii) Provide the name and structure for any monosubstituted benzene compound. **[3 Marks]**

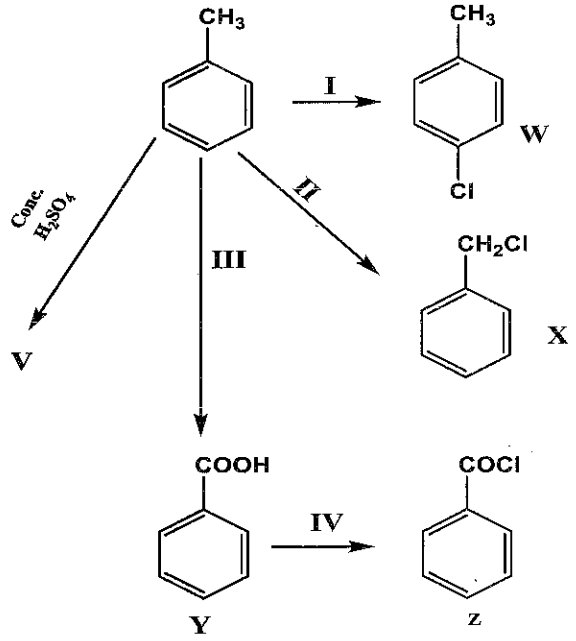
8. (a) Hydrocarbons with high molecular mass are subjected to a process known as cracking.

(i) Define the term cracking.

(ii) What conditions are used for cracking?

(iii) Differentiate between cracking and reforming. **[3 Marks]**

(b) The following scheme shows some reactions of toluene



(i) Suggest reagents and conditions for reaction I to IV.

(ii) Draw the structure of V.

(iii) Give the IUPAC name of V, W, Y and z. **[5 Marks]**

(c) **B** is a saturated unbranched hydrocarbon compound. When treated with chlorine under free radical conditions it produces a mixture of chlorinated compounds including **C**, $C_6H_{12}Cl_2$. Dehydrochlorination of **C** produces **D**, C_6H_{10}. When **D** is oxidized by hot conc. $KMnO_4$, two compounds, CO_2 and a carboxylic acid are formed in equimolar amounts.

(i) Suggest reagent and condition for the conversion of C to D.

(ii) Suggest a structure for compound D.

(iii) Deduce the structure of compound B. **[2 Marks]**